

FIT3152 Assignment 1: Predictors of Confidence in Social Organisations

World Values Survey Analysis — Focus Country: Armenia (ARM)

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1 Introduction

Generative AI was used in this assignment. This report analyses a student-specific subset of the World Values Survey (WVS) to examine how participant attributes relate to confidence in social organisations, and how those relationships differ between **Armenia** and **all other countries as a group**, both overall and over time. The report follows the assignment question order.

2 Data Preparation

The dataset was created exactly as required in the assignment specification using the student number as the random seed. Negative WVS special codes were recoded to **NA** because they represent non-substantive responses rather than valid low values.

3 Question 1 — Descriptive Analysis

3.1 1(a) Overall structure of the data

The final analysis file contains **100,000 rows** and **37 variables**: 3 metadata variables, 25 predictors, 8 confidence variables, and the derived grouping variable **Group**. Most variables are ordinal survey responses coded numerically, so they can be treated as numeric for this assignment. The main text fields are **Country** and **Group**, while the rest of the analysis is largely based on numeric variables.

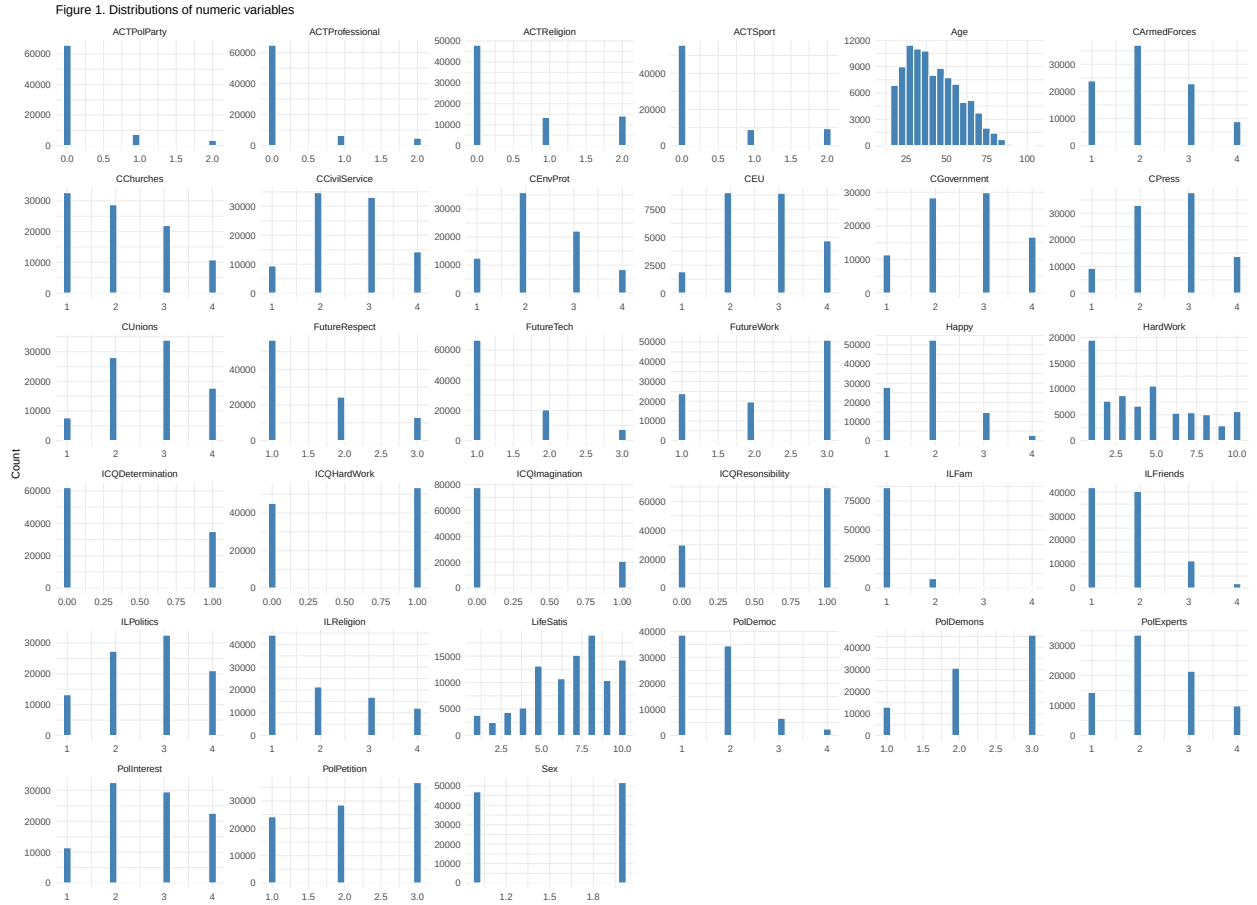
Table 1: Table 1. Dataset overview.

Metric	Value
Rows	100000
Columns	37
Countries	58
Wave range	1 to 7
Year range	1981 to 2023
Predictors	25
Confidence variables	8

The variables naturally fall into metadata (**Wave**, **Country**, **Year**), predictor attributes, confidence outcomes, and the derived grouping variable used for comparison. Because predictors vary widely in scale and meaning, later interpretation focuses on direction and relative strength rather than raw magnitudes alone.

3.2 1(b) Distribution of numerical and non-numerical responses

The numerical variables are measured on different scales. **Age** is continuous, **LifeSatis** and **HardWork** use broader numeric scales, and several **ICQ*** variables behave like binary indicators. This means the raw ranges are not directly comparable, so interpretation should focus on patterns, concentration, and direction rather than absolute magnitudes alone.



The distributions show that **ILFam** is heavily concentrated at the lowest score, **LifeSatis** and **HardWork** are widely spread, and many **ICQ*** variables behave like binary indicators. Most confidence variables cluster around the middle categories, suggesting moderate rather than highly polarised confidence. Non-numeric variety mainly comes from **Country** and the derived **Group** field, where the pooled “Other” category contains many distinct country codes and therefore masks substantial internal heterogeneity.

Table 2: Number of observations in Armenia and the pooled other-country group.

Group	n	Percent
Armenia	1203	1.2
Other	98797	98.8

3.3 1(c) Missing values and data quality issues

Missingness is substantial and uneven, so it is one of the most important descriptive features of the dataset. This is partly because some questions are not available in all countries or waves, and partly because WVS uses special negative codes for non-substantive answers.

Table 3: Variables with the most missing values after recoding special codes to NA.

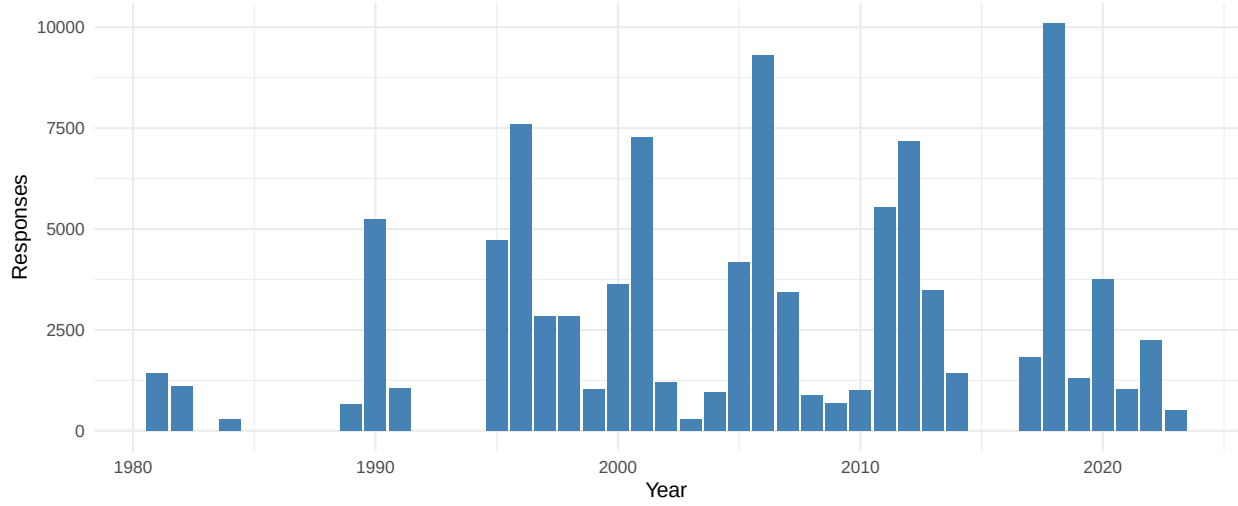
Variable	Missing	MissingPct
CEU	75504	75.5
ACTSport	26061	26.1
ACTReligion	24378	24.4
ACTProfessional	23865	23.9
ACTPolParty	23416	23.4
HardWork	22538	22.5
CEnvProt	21064	21.1
PolExperts	20650	20.6
PolDemoc	17782	17.8
CGovernment	13717	13.7
CUnions	12670	12.7
PolDemons	10886	10.9

This matters because summaries, tests, and correlations are not always based on the same sample size, especially in the over-time analysis. It also means that stronger or weaker relationships may partly reflect differing data availability across variables rather than only substantive differences.

3.4 1(d) Anything else of relevance or interest

Another important descriptive feature is the distribution of responses over time. Later years contribute many more observations than the earliest years, so temporal trends may partly reflect real attitudinal change and partly reflect changes in coverage, country participation, and sample size.

Figure 2. Number of survey responses by year



Overall, the dataset is large and informative but structurally uneven because of mixed scales, substantial missingness, imbalance between Armenia and the pooled Other group, and uneven wave coverage. These features do not prevent analysis, but they do mean later comparisons should be interpreted as approximate summaries rather than perfectly like-for-like contrasts.

4 Question 2 — Focus Country vs Other Countries (Ignoring Time)

All waves are pooled for Question 2, so the results below compare Armenia with the combined other-country group independent of time.

4.1 2(a) How do participant responses for Armenia differ from the other countries?

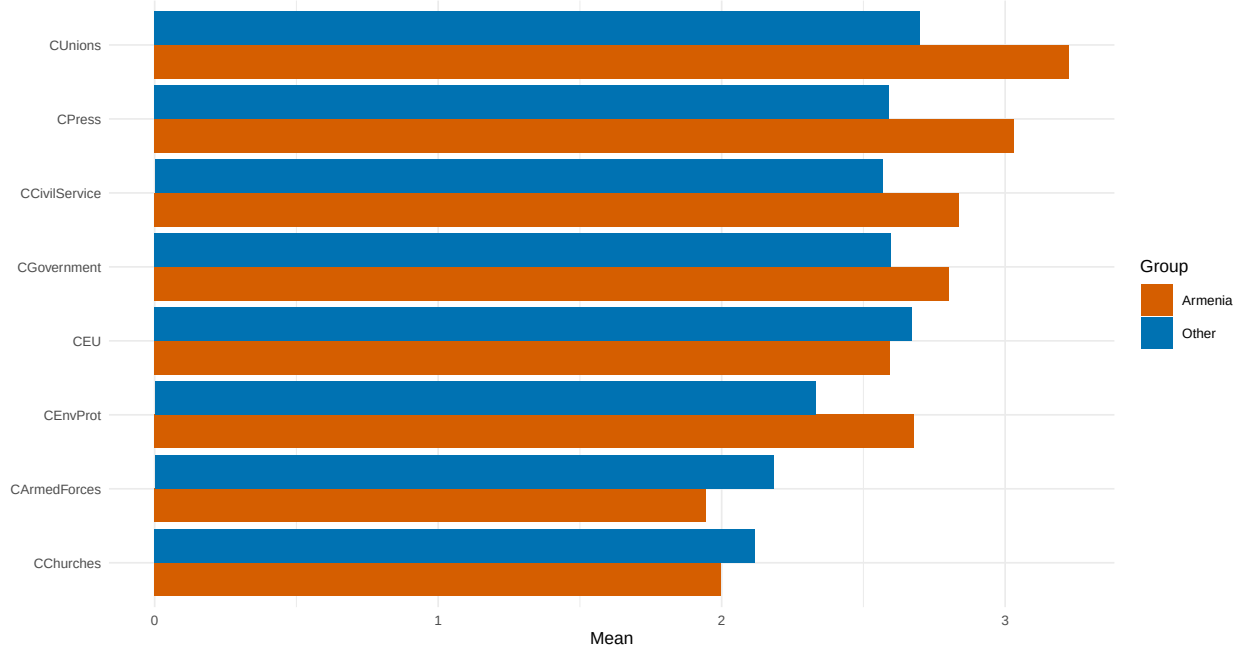
Group means were compared first, followed by Wilcoxon rank-sum tests for each variable. Wilcoxon tests were used because the variables are ordinal and the two groups are highly unequal in size.

Table 4: Largest Armenia-versus-Other differences across predictors and confidence variables.

Variable	ArmeniaMean	OtherMean	AbsDiff	PValue	Significant
Age	43.850	41.853	1.997	<0.001	TRUE
LifeSatis	5.295	6.787	1.492	<0.001	TRUE
CUnions	3.224	2.700	0.524	<0.001	TRUE
CPress	3.031	2.591	0.439	<0.001	TRUE
CEnvProt	2.678	2.331	0.347	<0.001	TRUE
ACTPolParty	0.504	0.182	0.322	<0.001	TRUE
ACTProfessional	0.526	0.207	0.319	<0.001	TRUE
PolPetition	2.450	2.136	0.314	<0.001	TRUE
CCivilService	2.837	2.567	0.270	<0.001	TRUE
ICQHardWork	0.803	0.539	0.263	<0.001	TRUE

Variable	ArmeniaMean	OtherMean	AbsDiff	PValue	Significant
CArmedForces	1.945	2.183	0.237	<0.001	TRUE
CGovernment	2.803	2.598	0.204	<0.001	TRUE

Figure 3. Mean confidence levels: Armenia vs other countries



Armenia differs from the other countries in several areas, especially well-being, religion, and some confidence variables. These differences are stronger in values and trust-related responses, rather than being spread across all variables equally.

4.2 2(b) How well do participant responses predict confidence in social organizations in Armenia?

Predictive quality was assessed using pairwise Pearson correlations between the 25 predictor variables and the 8 confidence variables. Because the assignment allows the ordinal variables to be treated as numeric, correlation is an appropriate first-pass measure of predictive strength. Larger absolute correlations indicate stronger predictive relationships.

Figure 4. Armenia: predictor–confidence correlations

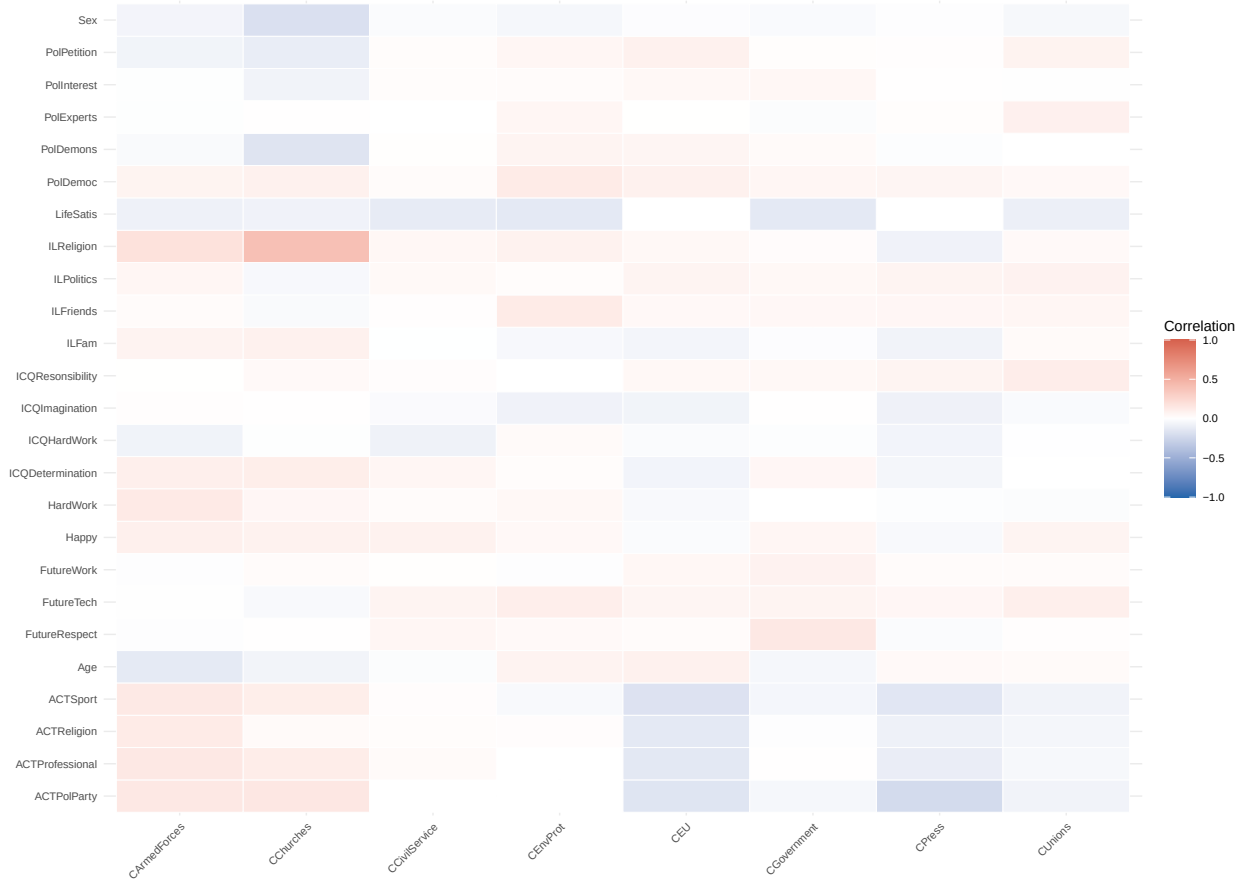


Table 5: Strongest predictor–confidence pairs in Armenia.

Predictor	Confidence	Correlation	AbsCorrelation
ILReligion	CChurches	0.397	0.397
ACTPolParty	CPress	-0.221	0.221
Sex	CChurches	-0.187	0.187
ILReligion	CArmedForces	0.182	0.182
ACTSport	CEU	-0.175	0.175
PolDemons	CChurches	-0.164	0.164
ACTPolParty	CEU	-0.160	0.160
ACTSport	CPress	-0.151	0.151
ACTPolParty	CChurches	0.148	0.148
FutureRespect	CGovernment	0.143	0.143

The Armenian correlations are mostly modest, so confidence is only partly predictable from the available attributes. The clearest pair is **ILReligion** → **CChurches**, showing that the importance of religion is closely associated with confidence in churches. More broadly, Armenia’s predictive structure is narrow, with only a small set of stronger relationships. In this report, “predictive quality” is interpreted as the strength of association rather than as a causal or out-of-sample forecasting model. Institutions with correlations near zero across most predictors are therefore treated as less

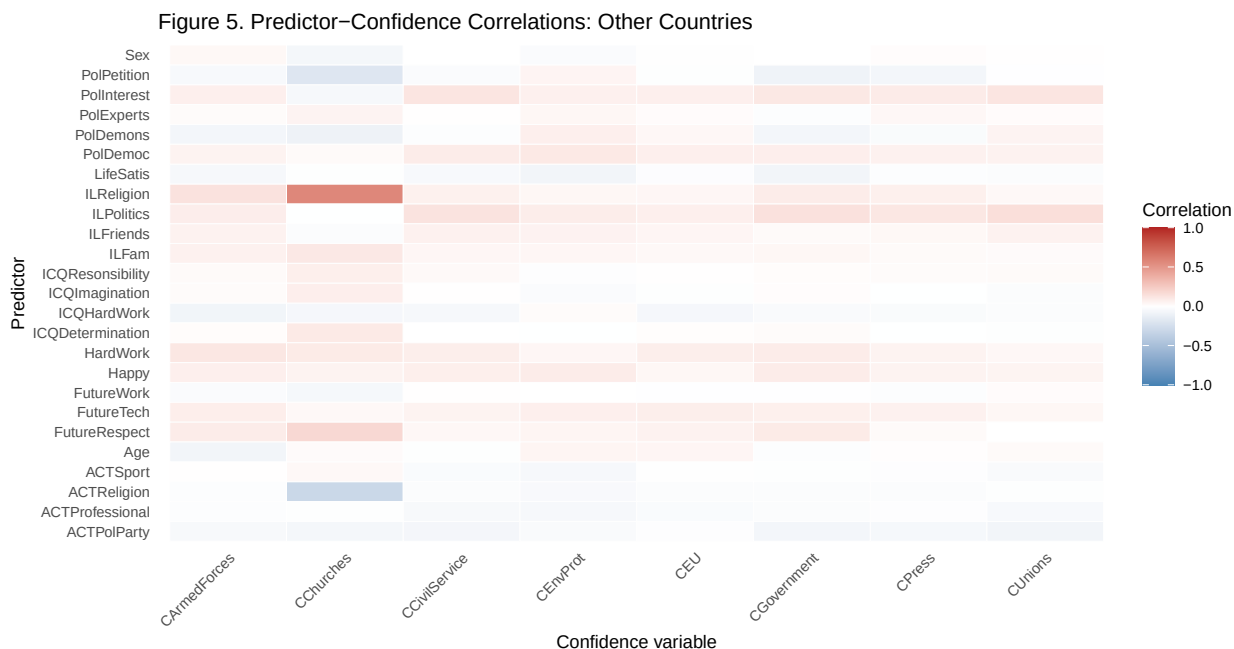
reliably predicted by the assigned attributes.

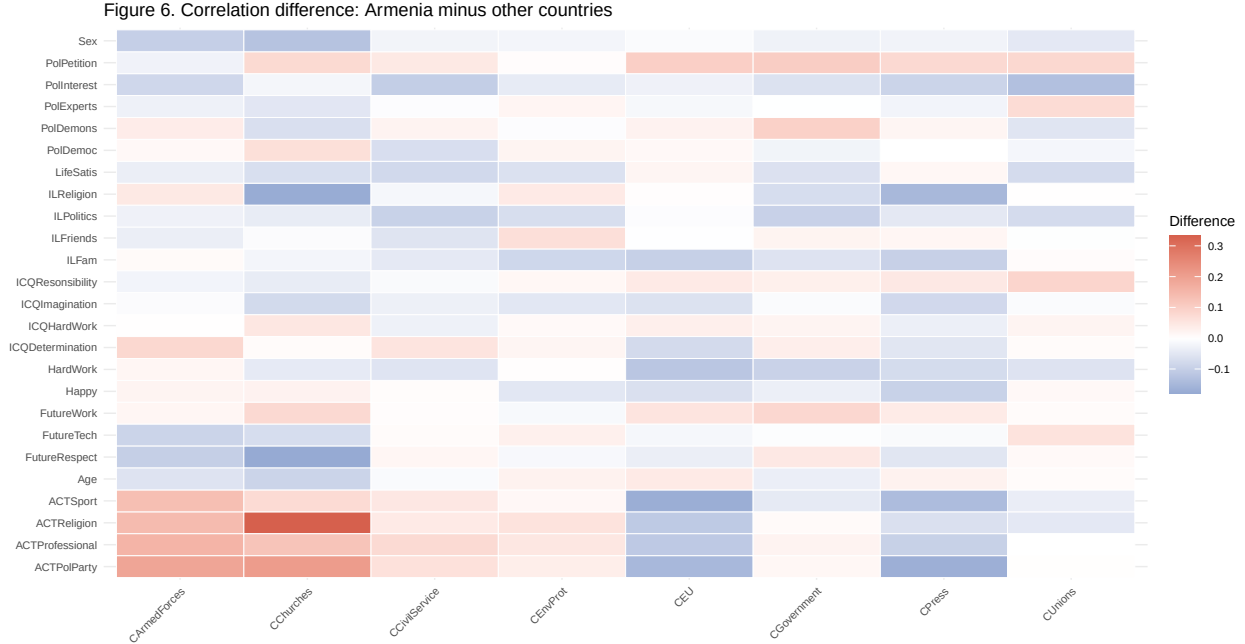
4.3 2(c) How do the other countries compare, and what are the strongest predictors there?

The same correlation analysis was repeated for the pooled other-country group.

Table 6: Strongest predictor–confidence pairs in the other-country group.

Predictor	Confidence	Correlation	AbsCorrelation
ILReligion	CChurches	0.570	0.570
ACTReligion	CChurches	-0.301	0.301
PolPetition	CChurches	-0.188	0.188
FutureRespect	CChurches	0.182	0.182
ILPolitics	CUnions	0.152	0.152
ILPolitics	CGovernment	0.139	0.139
ILReligion	CArmedForces	0.136	0.136
ILPolitics	CCivilService	0.133	0.133
PolInterest	CUnions	0.123	0.123
PolInterest	CCivilService	0.121	0.121





Overall, correlations in both groups are mostly weak. Armenia has a few more concentrated standout relationships, while the other-country group shows a more spread-out but still generally weak pattern. The difference heatmap mainly shows where the two groups differ, rather than indicating strong predictive power in either group.

5 Question 3 — Focus Country vs Other Countries Over Time

Wave was used instead of year because it provides a cleaner common temporal structure across countries and aligns better with the WVS collection design.

5.1 3(a) How do participant responses vary over time in Armenia and in the other-country group?

To study the effect of time, group means were calculated by wave and Kruskal–Wallis tests were used to assess whether Armenia showed significant differences across waves. Kruskal–Wallis was chosen because the survey variables are ordinal and the wave group sizes are unequal.

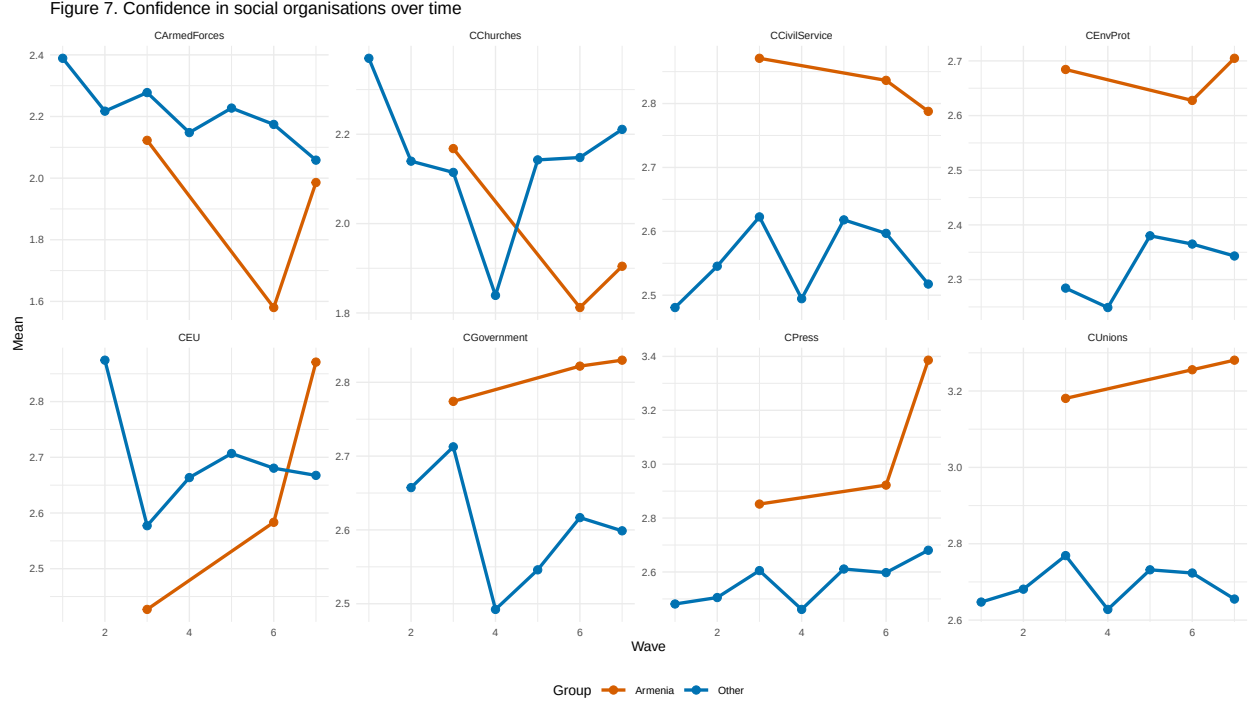


Table 7: Armenia variables with the strongest evidence of wave-to-wave change (Kruskal–Wallis tests).

Variable	PValue	Significant
ACTPolParty	1.32e-213	TRUE
ACTProfessional	1.88e-197	TRUE
ACTSport	2.22e-195	TRUE
ACTReligion	5.81e-159	TRUE
Happy	1.07e-43	TRUE
ILReligion	2.70e-42	TRUE
ICQDetermination	2.21e-30	TRUE
LifeSatis	3.44e-29	TRUE
PolDemons	1.02e-25	TRUE
CPress	1.92e-24	TRUE

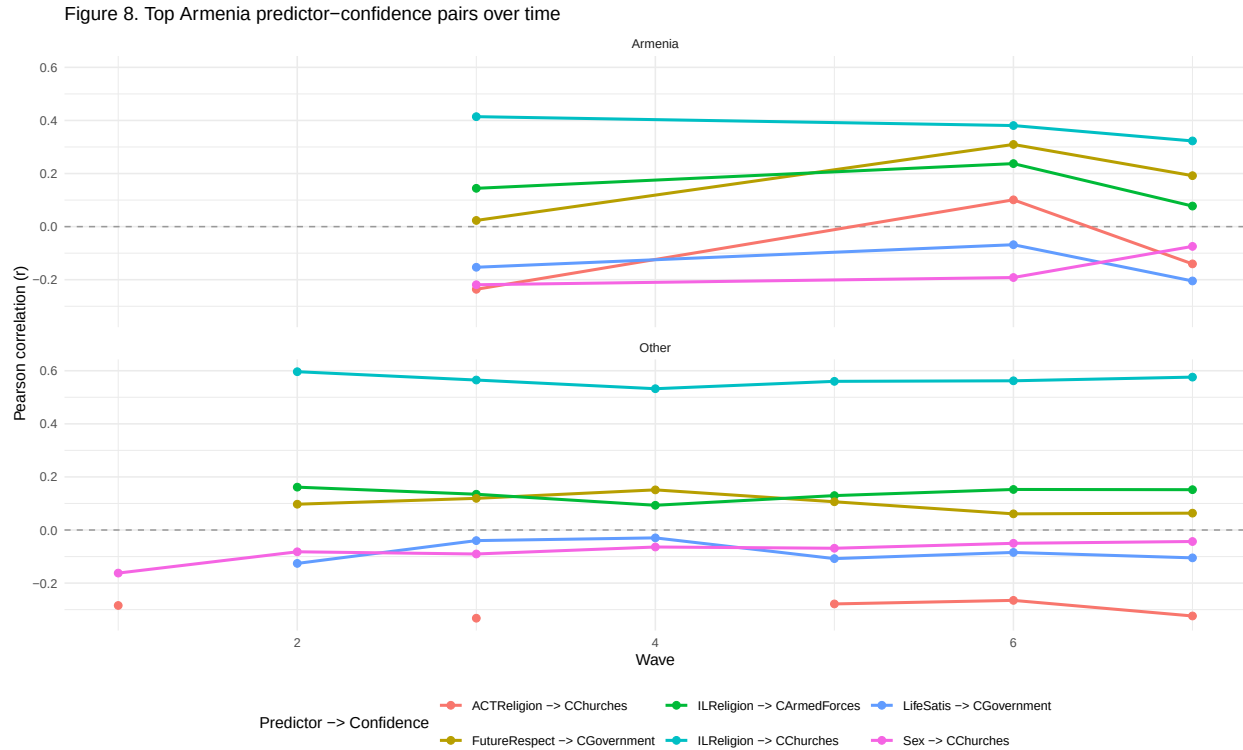
Armenia has a more limited time profile because it appears in fewer waves than the pooled Other-country group, so its trend is based on fewer observed points. Even so, several Armenian variables change clearly across waves, and the Kruskal–Wallis results show that many of these changes are statistically significant. The strongest wave-to-wave changes in Armenia are mainly found in participation and attitudinal variables, although some confidence measures, such as confidence in the press, also change over time. The pooled Other-country group does not stay flat, but its pattern is more averaged because it combines many countries, which reduces some country-specific swings. So the main difference is not simply the level of responses, but the way the trends unfold over time: Armenia often shows clearer directional shifts, while the pooled group reflects a broader combined pattern. The faceted line plot was chosen because it makes it easy to compare these trajectories across both groups and across the different confidence variables.

5.2 3(b) How does the ability of participant responses to predict confidence change over time?

To answer this question, correlations between predictors and confidence variables were recalculated separately within each wave for Armenia and for the pooled Other group. This makes it possible to see whether the strongest predictors remain stable or whether their predictive importance changes over time.

Table 8: Six strongest Armenia predictor–confidence pairs across waves.

Predictor	Confidence	mean_abs_cor
ILReligion	CChurches	0.373
FutureRespect	CGovernment	0.175
Sex	CChurches	0.162
ACTReligion	CChurches	0.159
ILReligion	CArmedForces	0.153
LifeSatis	CGovernment	0.142



The strongest Armenian relationships remain concentrated in a small subset of predictors, especially religion-related attitudes. **ILReligion → CChurches** remains the clearest example across waves. Some Armenian correlations stay fairly stable, while others change in strength over time, so predictor importance is not fixed. The pooled Other-country group shows a broader but generally more moderated pattern. Figure 8 focuses on the six strongest Armenian pairs, making it easier to compare how these key relationships evolve over time in both groups.

6 Conclusion

Armenia differs from the pooled Other-country group both descriptively and predictively. The dataset is large and useful but uneven in scale, missingness, and time coverage. Predictively, Armenia is characterised by a narrower set of stronger relationships, especially religion-related attitudes and confidence in churches. Over time, Armenia also shows a more compressed profile because it appears in fewer waves. The main limitations are uneven missingness, unequal country-wave coverage, the loss of country-specific detail inside the pooled Other group, and the fact that correlations capture association rather than causation. Even with these limitations, the analysis gives a coherent answer to all three research questions and shows where Armenia is most distinct from the broader international pattern.

7 Appendix — R Code

The main report keeps code visibility limited to preserve readability, but the assignment also requires the R code to be included. The core code is therefore reproduced below in machine-readable form.

```
set.seed(34107029)
VCData <- read.csv("WVSEExtract.csv")
VC <- VCData[sample(1:nrow(VCData), 100000, replace = FALSE), ]
VC <- VC[, c(1:3,
             sort(sample(4:50, 25, replace = FALSE)),
             sort(sample(51:65, 8, replace = FALSE)))]

country_data <- VC %>%
  mutate(Group = ifelse(Country == "ARM", "Armenia", "Other"))

conf_cols <- names(country_data)[grepl("^C", names(country_data)) & sapply(country_data, is.numeric)]
pred_cols <- setdiff(names(country_data)[sapply(country_data, is.numeric)], c("Wave", "Year", "Country"))
analysis_cols <- c(pred_cols, conf_cols)

country_data <- country_data %>%
  mutate(across(all_of(analysis_cols), ~ ifelse(.x < 0, NA, .x)))

armenia_data <- country_data %>% filter(Group == "Armenia")
other_data <- country_data %>% filter(Group == "Other")
compare_groups <- function(var) {
  sub <- country_data %>% select(Group, all_of(var)) %>% filter(!is.na(.data[[var]]))
  test <- suppressWarnings(wilcox.test(sub[[var]] ~ sub$Group))
  tibble(
    Variable = var,
    ArmeniaMean = mean(sub[[var]][sub$Group == "Armenia"], na.rm = TRUE),
    OtherMean = mean(sub[[var]][sub$Group == "Other"], na.rm = TRUE),
    PValue = test$p.value
  )
}

q2a_tests <- map_dfr(c(pred_cols, conf_cols), compare_groups) %>%
```

```

mutate(AbsDiff = abs(ArmeniaMean - OtherMean), Significant = PValue < 0.05) %>%
  arrange(desc(AbsDiff))
cor_arm <- cor(armenia_data[, pred_cols], armenia_data[, conf_cols], use = "pairwise.complete.obs")
cor_oth <- cor(other_data[, pred_cols], other_data[, conf_cols], use = "pairwise.complete.obs")

arm_pairs <- as.data.frame(cor_arm) %>%
  rownames_to_column("Predictor") %>%
  pivot_longer(-Predictor, names_to = "Confidence", values_to = "Correlation") %>%
  mutate(AbsCorrelation = abs(Correlation)) %>%
  arrange(desc(AbsCorrelation))

oth_pairs <- as.data.frame(cor_oth) %>%
  rownames_to_column("Predictor") %>%
  pivot_longer(-Predictor, names_to = "Confidence", values_to = "Correlation") %>%
  mutate(AbsCorrelation = abs(Correlation)) %>%
  arrange(desc(AbsCorrelation))

cor_diff <- cor_arm - cor_oth
time_means <- country_data %>%
  group_by(Group, Wave) %>%
  summarise(across(all_of(c(pred_cols, conf_cols)), ~ mean(.x, na.rm = TRUE)), .groups = "drop")
  pivot_longer(cols = -c(Group, Wave), names_to = "Variable", values_to = "Mean")

kw_results <- map_dfr(c(pred_cols, conf_cols), function(col) {
  vals <- country_data[[col]][country_data$Group == "Armenia"]
  waves <- country_data$Wave[country_data$Group == "Armenia"]
  keep <- !is.na(vals) & !is.na(waves)
  test <- kruskal.test(vals[keep] ~ as.factor(waves[keep]))
  tibble(Variable = col, PValue = test$p.value, Significant = test$p.value < 0.05)
}) %>% arrange(PValue)

compute_cor_over_time <- function(data, group_name) {
  waves <- sort(unique(data$Wave))
  bind_rows(lapply(waves, function(w) {
    sub <- data %>% filter(Group == group_name, Wave == w)
    if (nrow(sub) < 10) return(NULL)
    cor_mat <- tryCatch(cor(sub[, pred_cols], sub[, conf_cols], use = "pairwise.complete.obs"),
      error = function(e) NULL)
    if (is.null(cor_mat)) return(NULL)
    as.data.frame(cor_mat) %>%
      rownames_to_column("Predictor") %>%
      pivot_longer(-Predictor, names_to = "Confidence", values_to = "Correlation") %>%
      mutate(Wave = w, Group = group_name)
  }))
}

cor_time_arm <- compute_cor_over_time(country_data, "Armenia")
cor_time_oth <- compute_cor_over_time(country_data, "Other")

```

```

cor_time_all <- bind_rows(cor_time_arm, cor_time_oth)

top_pairs_arm <- cor_time_arm %>%
  group_by(Predictor, Confidence) %>%
  summarise(mean_abs_cor = mean(abs(Correlation), na.rm = TRUE), .groups = "drop") %>%
  arrange(desc(mean_abs_cor)) %>%
  slice_head(n = 6)

top_pair_keys <- paste(top_pairs_arm$Predictor, top_pairs_arm$Confidence, sep = "_")

```

8 Appendix — Generative AI Use

Item	Detail
Technology	ChatGPT and Claude Code
Information generated	R code fragments for reshaping data, correlations, tests, and graphics
Prompts used	Requests for correlation heatmaps, comparison plots, wave-based analysis, and R Markdown restructuring
How output was used	Output was reviewed, edited, reorganised, and integrated into the final workflow